MILITARY SPECIFICATION

DISPENSER, BOMB, TYPE SUU-21/A

1. SCOPE

1.1 This specification covers the requirements for one type of equipment designated as Dispenser, Bomb, Type SUU-21/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	Plastic Film (Polyethylene Thin Gage)
PPP-T-60	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging

Military

MIL-P-116	Preservation, Methods Of
MIL-W-5088	Wiring, Aircraft, Installation Of
MIL-F-7179	Finishes And Coatings: General Specification For Protection Of Aerospace Weapons Systems, Structures And Parts
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-C-25731	Crates, Wood, For Lightweight, Bulky Aircraft Items
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards Order Of Precedence For The Selection Of
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods For Aerospace And Ground Equipment

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MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS3103	Connectors-Receptacle, Electric, For Potting
MS33586	Metals, Definition Of Dissimilar

DRAWINGS

Air Force

57J46345	Wiring Diagram, Bomb Release System, SUU-21/A Dispenser
62D34085	Tester Assembly, Dispenser, SUU-21/A
63D34060	Dispenser, Bomb, Type SUU-21/A Assembly Of
63E36800	Relay Box, SUU-21/A Dispenser
64C34567	Cable Assy, SUU-21/A Testing

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

438	AGE Controls For Synthetic Rubber Parts

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL AEROSPACE STANDARDS COMMITTEE

NAS1400	Rivet-Blind, Self-Plugging, Mechanically Locked Spindle

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, N.W., Washington, D.C. 20036.)

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3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Components. The SUU-21/A Dispenser shall consist of: a nose cone assembly; a tail cone assembly; an attachment kit; a center section assembly which includes a structure assembly, an electrical harness installation, a relay box assembly, an ejector installation, a linkage installation; and a door installation.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Rubber parts. All synthetic rubber parts shall meet the requirements of ANA Bul 438.

3.4.2 Rivets. All aluminum alloy blind rivets shall conform to NAS1400.

3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.5 Design and construction.

3.5.1 Design. The SUU-21/A Dispenser shall be designed as shown on drawing 63D34060. Care shall be taken to maintain the design requirements incorporated therein. Provisions shall be included for mounting lugs in three different positions so that the dispenser can be attached to bomb racks with 14 inch, 20 inch and 30 inch suspension capabilities. The design of the dispenser shall be such that it is capable of dispensing BDU-33/B or MK-106 Mod. 1 Practice Bombs for subsonic, transonic and supersonic aircraft, one at a time, by each successive electrical bomb release signal.

3.5.1.1 The dispenser shall include six (6) ejector mechanisms to obtain proper ejection separation for each BDU-33/B or MK-106 Mod. 1 Practice Bomb.

3.5.1.2 Relay box. A relay box shall be provided in the dispenser. The major function of the relay box shall be to energize sequentially the release solenoids for each practice bomb.

3.5.1.2.1 Selective door operation model. The selective relay box shall have a ground option of manual or automatic. The doors shall close automatically after each release if the automatic mode is chosen. In the manual mode the doors must be operated electrically from the cockpit and do not close automatically after releases.

3.5.1.2.2 Nonselective door operation model. The nonselective relay box shall automatically close the dispenser doors after each bomb release.

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3.5.1.3 Door actuator and linkage drive. An electric linear actuator shall be provided to operate the doors of the dispenser through a linkage drive. The actuator shall have ample capacity to assure positive opening and closing of doors under all flight conditions generated by carrying aircraft. The linkage drive shall control a limit switch to isolate a bomb release signal until the doors are fully opened.

3.5.1.4 Associated electrical wiring and harnesses. Electrical wiring and harnesses shall be as shown on drawing 57J46345 and its associated detail drawings. All wiring shall meet the requirements of MIL-W-5088.

3.5.1.5 Trainer, weapons simulator. There shall be provisions for installing a weapons simulator in the tail cone assembly. A part of the tail section shall be removable to provide access for resetting of its cams.

3.6 Construction. The dispenser shall be constructed to comply with all the requirements set forth herein and with the requirements of the engineering drawings and military specifications. The dispenser construction shall be semi-monocoque consisting of aluminum alloy sheet reinforced locally with stringers and frames. Bulkheads shall be incorporated to support the dispensing mechanisms. A hardback casting shall be provided to take swaybrace loads and ejection loads and to provide for installation of the mounting lugs. All screw threads shall be in accordance with MIL-S-7742.

3.7 Performance. The SUU-21/A Dispenser shall operate satisfactorily under the following conditions:

3.7.1 Shock. The dispenser shall operate satisfactorily after being subjected to the application of impact shocks of 15g with each shock impulse having a duration of 11 ± 1 milliseconds, without the use of shock insulators.

3.7.2 Vibration. The dispenser shall operate satisfactorily while being subjected to vibration as specified in 4.6.5.

3.7.3 Acceleration. The dispenser shall operate satisfactorily after being subjected to an 8g downward acceleration, stabilized and maintained for one minute. The dispenser shall then be rotated 90° about the longitudinal (roll) axis and shall operate satisfactorily after being subjected to an 8g sideward acceleration, stabilized and maintained for one minute.

3.7.4 Sand and dust. The dispenser shall operate satisfactorily after prolonged exposure to desert atmospheres containing sand and dust particles.

3.7.5 Humidity. The dispenser shall operate satisfactorily for a period of 240 hours at 95 percent relative humidity in temperatures ranging from 20° Centigrade (C) to 71°C (68° Fahrenheit (F) to 160°F).

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3.7.6 Altitude. The dispenser shall operate satisfactorily at an altitude of 50,000 feet with a barometric pressure of 3.44 inches of mercury and an ambient temperature of -54°C (-65°F).

3.7.7 High temperature. The dispenser shall operate satisfactorily during and after subjection to temperatures up to and including 71°C (160°F).

3.7.8 Low temperature. The dispenser shall be capable of withstanding storage conditions at a temperature of -62°C (-80°F) and shall operate satisfactorily while being subjected to temperatures of -54°C (-65°F).

3.7.9 Temperature shock. The dispenser shall operate satisfactorily during and after subjection to variations of temperatures from 85°C (185°F) to -40°C (-40°F).

3.7.10 Salt spray. The dispenser shall show no signs of corrosion which could adversely affect its operation, and shall operate satisfactorily after prolonged exposure in an area near a large body of salt water where salt spray is generally encountered in the atmosphere.

3.7.11 Fungus. The dispenser shall operate satisfactorily after prolonged exposure to tropical atmospheres containing fungus spores.

3.7.12 Rated electrical load. All electrical components of each dispenser shall operate properly under rated load conditions and in conjunction with the aircraft delivery system.

3.8 Dimensions. The dispenser shall be of an aerodynamic shape of approximately 156.25 inches in length with a maximum diameter of 17 inches.

3.9 Weight. The weight of the dispenser, without bombs, shall be approximately 470 pounds.

3.10 Finish. The finish of the dispenser shall be in accordance with MIL-F-7179.

3.11 Identification of product. Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Government furnished property. When the contract or purchase order so provides, the Government will furnish the following items to the contractor, upon his request, for accomplishment of specified functional tests.

ITEM NO.	STOCK NO.	ITEM DESCRIPTION	QUANTITY
1	1190-022-5849CM	SUU-21/A Dispenser Tester	1 each
2	1190-867-7351CM	Cable Assembly	1 each

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3.12.1 Additional equipment. Testers, cables and procedures for checking out the two relay box configurations are available.

3.13 Government loaned property. The following items of equipment shall be loaned to the contractor, upon his request, as Government loaned property, for the purpose of conducting the functional test of the SUU-21/A Dispenser:

ITEM NO.	STOCK NO.	ITEM DESCRIPTION	QUANTITY
1	1190-566-1902CM or 1190-074-2640CM	Loading Device, Practice Bomb, Type MD-1 Loading Device, Practice Bomb, Type HLU-129/E	1 each 1 each
2	1190-022-5849CM	SUU-21/A Dispenser Tester	1 each
3	1325-529-7566	Bomb Practice, 25 Pound BDU-33/B	6 each
4	1115-724-3772CM	Bomb, Practice, Type MK-106 Mod 1	6 each

3.14 Workmanship. The SUU-21/A Dispenser shall be fabricated and finished in a manner such that criteria of appearance, fit, and adherence to specified tolerances shall be observed. These criteria must apply equally to all accessories and to all external nonfunctioning parts (e.g., cases, cover plates, and nameplates) as well as to functional parts. Particular attention shall be given to neatness, marking of parts and assemblies, plating, painting, welding, machine screw assemblage, freedom of parts from burrs and sharp edges which might cause injury to personnel or damage to associated equipment, and conformance to drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the Dispenser, Bomb, Type SUU-21/A shall be classified as follows:

a. Preproduction tests. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove that the dispenser meets the requirements of this specification.

b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

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4.3 Test conditions. The test conditions are described under the individual test to which they apply.

4.4 Preproduction testing.

4.4.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.4.1.1 In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-STD-831 and furnish the report to the procuring activity.

4.4.1.2 All test equipment used in the preproduction tests shall be calibrated per MIL-C-45662.

4.4.2 Preproduction tests. Preproduction tests shall consist of all applicable tests described under 4.6.

4.5 Acceptance tests. Acceptance test shall consist of individual tests.

4.5.1 Individual tests. Each dispenser shall be subjected to the following:

a. Examination of product per 4.6.1.

b. Conformance to drawings per 4.6.2.

c. Functional Tests per 4.6.3.1 items a thru l only or 4.6.3.2 items a thru o only.

4.6 Test methods.

4.6.1 Examination of product. Each dispenser shall be examined to determine that the workmanship meets the requirements of this specification.

4.6.2 Conformance to drawings. Each dispenser shall be inspected for strict conformance to the pertinent drawings and specifications.

4.6.3 Functional test. A functional test shall be performed on each sample selected to insure proper operation of the loading and ejection mechanisms and their components.

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4.6.3.1 The functional test of dispensers with nonselective relay boxes shall be performed as follows:

a. Install all equipment in the dispenser including the ejectors, the relay box, and the pads and adapters for the MK-106 Mod 1 Practice Bomb. Connect cable assembly, part number (PN) 64C34567, to the dispenser tester, PN 62D34085, and to connector, MS3103-32-7-P, (FWD) of the dispenser.

b. Check that all switches on the tester are in the "OFF" position and connect a source of 28 volts direct current (vdc) power to tester input receptacle.

c. Turn on power to tester by turning on "PWR" switch. "PWR-ON" light shall illuminate. Check that all press-to-test lights illuminate when depressed.

d. Turn on switch "S1." Dispenser doors shall open, indicator DS-11 on tester shall blink while the doors are in the process of opening and illuminate steady when the doors are fully opened. Turn off switch "S1." DS-11 shall extinguish.

e. Attach the MD-1 or HLU-129E loading device to side of dispenser and load six practice bombs. Simulated MK-106 practice bombs of proper weight and diameter may be used.

f. After the six bombs have been loaded, set the indicator on the relay box to the No. 1 position. Remove the loading device from the dispenser and place boxes filled with sawdust below the dispenser as receptacles for the ejected bombs.

g. Turn switch "S1" on for approximately 20 seconds (DS-11 illuminates). Turn switch "S1" "OFF" then back on immediately. The doors shall close, indicator DS-11 shall blink while the doors are in the process of closing and extinguish when the doors are fully closed. Turn off "S1" and wait at least 30 seconds.

h. Turn "S1." The doors will open. Indicator DS-11 will function as above during dispenser door opening or closing.

i. Momentarily turn on switch "S3." The bomb must be ejected from the No. 1 position, the dispenser doors must close automatically, and the indicator on the relay box must automatically advance to position No. 2. Indicator DS-11 shall blink while the doors are closing, and extinguish when the doors are closed. Turn off switch "S1."

j. Wait approximately 30 seconds then turn on "S1" to open dispenser doors. Repeat paragraph i for bomb positions No. 2 and No. 3.

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k. While dispenser doors are open turn off all switches on tester, remove the "aft" cover plate and adapter harness from the dispenser. Disconnect cable assembly 64C34567 from the forward connector and reconnect it to the aft connector (MS3103-32-7-P).

l. Repeat paragraph h, i, and j for bomb positions No. 3 through No. 6.

m. Change the adapters in the dispenser to those for the BDU-33/B practice bomb. Repeat the test using 6 each BDU-33/B practice bombs.

n. The dispenser is acceptable if no malfunction occurs throughout the test.

4.6.3.2 The functional test of dispensers with selective option relay boxes shall be performed as follows:

a. Install all equipment in the dispenser, including the ejectors, the relay box, and the pads and adapters for the MK-106 Mod 1, Practice Bomb. Connect cable assembly, PN 64C34567, to the SUU-21/A dispenser tester, PN 62D34085, and to connector, MS3103-32-7-P, (FWD) of the dispenser.

b. Check that all switches on the tester are in the "OFF" position, and connect a source of 28 vdc power to tester input receptacle.

c. Turn on power to tester by turning on "PWR" switch. "PWR-ON" shall illuminate. Check that all press-to-test lights illuminate when depressed.

d. Turn on switch "S1." Dispenser doors shall open, indicator DS-11 on tester shall blink while the doors are in the process of opening, and illuminate steady when the doors are fully open. Turn off "PWR" switch, and switch "S1."

e. Attach the MD-1 or HLU-129/E loading device to the side of the dispenser and load six practice bombs or simulated practice bombs.

f. After the bombs have been loaded, set the bomb selector switch on the relay box to the No. 1 position. Place mode toggle switch located on the relay box to the "automatic" position. Remove the loading device from the dispenser and place boxes filled with sawdust below the dispenser as receptacles for the ejected bombs.

g. Turn "PWR" switch and switch "S1" back on. "PWR-ON" light and indicator DS-11 shall illuminate.

h. Momentarily turn on switch "S3." The bomb must be ejected from the No. 1 position, the doors must close automatically, selector switch knob on relay box must automatically advance to position No. 2, and indicator DS-11 shall blink while the doors are in the process of closing, and extinguish when the doors are fully closed.

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i. Turn "OFF" switch "S1" then back on. Dispenser doors shall open. Indicator DS-11 shall function as above during door opening and closing.

j. Repeat paragraphs h and i above for bomb positions No. 2 and No. 3.

k. While dispenser doors are open, turn off "PWR" switch and place mode toggle switch on dispenser relay box to the "manual" position. Remove "aft" cover plate. Disconnect cable assembly from "FWD" connector, MS3103-32-7-P, and reconnect it to the "aft" connector, MS3103-32-7-P.

l. Turn on "PWR" switch and switch "S1."

m. Release bombs from positions No. 4 through No. 6 by momentarily turning on switch "S3." A bomb shall be ejected each time "S3" is turned on. The dispenser doors shall not close after a release.

n. Turn off switch "S1." Indicator DS-11 shall extinguish.

o. Turn "S4" to the "ON" position. The dispenser doors shall close. Indicator DS-11 shall blink while the doors are closing, and shall extinguish when the doors are closed.

p. Change the pads and adapters in the dispenser to those for BDU-33/B Practice Bombs. Repeat the test using six each BDU-33/B Practice Bombs.

q. The dispenser is acceptable if no malfunction occurs throughout the testing.

4.6.3.3 Optional relay box test procedure for Relay Box 63E36800-7. This procedure is provided so that the Relay Box may be tested using the dispenser tester, PN 62D34085, before being installed on the dispenser. Test procedures of 4.6.3.2 must be accomplished regardless of results of this test.

a. Check that all switches on the tester are in the "OFF" position and circuit breaker on relay box is pushed in. Place relay box toggle switch in automatic position.

b. Connect cable assembly, 64D33386, between unit under test and tester; connect a source of 28 vdc to tester input receptacle.

c. Turn on power to tester by turning on "PWR" switch. "PWR-ON" light shall illuminate.

d. Check that all press-to-test lights illuminate when depressed.

e. Close switch "S1." Indicator lights DS-10 and DS-12 shall illuminate.

f. Rotate relay box knob to the number 1 position.

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g. Momentarily turn on switch "S3," then release. Indicator DS-1 shall illuminate when "S3" is on, Indicator DS-10 shall extinguish, Indicator DS-11 shall illuminate and Indicator DS-12 shall remain illuminated.
NOTE: Do not hold "S3" in the "ON" position in excess of 10 seconds.

h. Momentarily hold "S2" in the "ON" position. Indicator DS-10 shall illuminate with DS-11 and DS-12 remaining illuminated. Release "S2." DS-10 shall extinguish.

i. Turn off switch "S1" and then turn back on. Indicators DS-10 and DS-12 shall illuminate. DS-11 shall extinguish.

j. Repeat steps g thru i above checking that Indicators DS-2 thru DS-6 illuminate when "S3" is momentarily held in the "ON" position.

k. Turn off all switches on tester.

l. Place relay box toggle switch to the manual position and turn on power switch. "PWR-ON" light shall illuminate.

m. Close switch "S1." Indicator lights DS-10 and DS-12 shall illuminate.

n. Rotate relay box knob to the number 1 position.

o. Momentarily hold "S3" in the "ON" position, then release. Indicator DS-1 shall illuminate momentarily with DS-10 and DS-12 remaining illuminated.

p. Repeat step o, until DS-2 thru DS-6 illuminate when "S3" is momentarily held in the "ON" position.

q. Momentarily hold "S4" in the "ON" position. Indicator DS-11 shall illuminate with DS-10 and DS-12 remaining illuminated. Release "S4." DS-11 shall extinguish.

r. Turn off all switches and disconnect the unit under test. Satisfactory completion of the above tests indicates an acceptable relay box.

4.6.3.4 Optional relay box test procedure for Relay Box 63E36800-5. This procedure is provided so that the Relay Box may be tested using the Dispenser Tester, PN 62D34085, prior to installation in the dispenser. Test procedures of 4.6.3.1 must be accomplished regardless of results of this test.

a. Check that all switches on the tester are in the "OFF" position and circuit breaker on relay box is pushed in.

b. Connect a source 28 vdc power to tester input receptacle and connect cable assembly 62D34590 to relay box receptacles and tester receptacle.

c. Turn on power to tester by turning on "PWR" switch. "PWR-ON" light shall illuminate.

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d. Check that all press-to-test lights illuminate when depressed.

e. Turn on switch "S1." Indicator lights DS-11 and DS-12 shall illuminate.

f. Momentarily hold switch "S2" in the "ON" position. Indicator DS-11 shall extinguish with DS-12 remaining illuminated. Indicator DS-10 shall illuminate. Release "S2." Indicator DS-10 shall extinguish and DS-11 and DS-12 shall illuminate.

g. Rotate stepping switch index knob, located on relay box, to the number 1 position.

h. Momentarily hold switch "S3" in the "ON" position. Indicators DS-1 and DS-11 shall illuminate. Indicator DS-12 shall extinguish. Release "S3." Indicator DS-1 shall extinguish and the stepping switch index knob on relay box shall automatically advance to the next position.
NOTE: Do not hold "S3" in the "ON" position in excess of 10 seconds.

i. Turn off switch "S1" and wait approximately 25 seconds, then turn "S1" back on. Indicators DS-11 and DS-12 shall illuminate when "S1" is returned to the "ON" position.

j. Repeat steps h and i checking that indicators DS-2 thru DS-6 illuminate when "S3" is momentarily held "ON."

k. Immediately after the illumination and extinguishing of DS-6, turn "S1" off then back on immediately. Indicator DS-11 shall illuminate.
NOTE: Switch "S1" must be returned back on within 10 seconds after being turned off.

l. Momentarily hold switch "S4" in the "ON" position. Indicator light DS-12 shall illuminate with DS-11 remaining illuminated.

m. Turn off all switches and wait at least 30 seconds before performing test of 3 seconds time delay Relay (K-103).

n. To test the 3 second time delay Relay (K-103), turn on power switch. "PWR-ON" light shall illuminate. While holding switch "S3" in the "ON" position, turn on switch "S1." Indicator light corresponding to relay box knob setting shall illuminate. Indicator DS-11 shall illuminate. Indicator DS-12 shall illuminate for approximately 3 seconds then extinguish.

o. The relay box is satisfactory if no malfunctions occurred throughout the test.

4.6.3.5 The dispensers shall be inspected for failures and functionally tested in accordance with 4.6.3.1 steps "g" thru "l" (for dispensers with nonselective relay boxes) and in accordance with 4.6.3.2 steps "g" thru "o" (for dispensers with selective option relay boxes) during or after each test, as specified in 4.6.4, 4.6.5, 4.6.6 and 4.6.7. BDU-33/B Practice Bombs may be used.

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4.6.4 Shock test. The shock test of a completely assembled and loaded dispenser shall be in accordance with Method 516, procedure III of MIL-STD-810 with the following exceptions.

a. The shock shall be applied without employing shock insulators.

b. The dispenser need not be operated during conditions of shock, but shall be operated after being subjected to each condition of shock.

4.6.4.1 The dispenser shall be loaded with 6 each BDU-33/B Practice Bombs during the shock test.

4.6.4.2 At the conclusion of the shock test, the equipment shall be examined for failures and operated in accordance with 4.6.3.5.

4.6.5 Vibration test. The dispenser, while loaded with 6 each BDU-33/B Practice Bombs, shall be vibrated without vibration isolators to produce a double amplitude of 0.032 plus 0.010 minus 0.000 inch when vibrated between 10 and 34 cycles per second.

4.6.5.1 Resonance. Resonant frequencies of the test specimen shall be determined by varying the frequency of applied vibration slowly through the range of 10 to 34 cycles per second. This test shall take a minimum of 60 minutes at room temperature. These periods of vibration shall be accomplished with vibration applied along each of three mutually perpendicular axes. If resonance is encountered, the dispenser shall be vibrated for 10 hours along the axis most likely to produce failure.

4.6.5.2 Nonresonance. If resonance is not encountered, the dispenser shall be vibrated for 25 hours at 33 cycles per second along the longitudinal axis.

4.6.6 Acceleration test. The acceleration tests of a completely assembled and loaded dispenser shall be in accordance with Method 513, procedures I & II of MIL-STD-810. The dispenser shall be loaded with 6 each BDU-33/B Practice Bombs during the acceleration test. To simulate down loading, the dispenser shall be mounted and locked to the centrifuge with the longitudinal (roll) axis normal to the arm of the centrifuge and the yaw axis parallel to the arm of the centrifuge. The forward end of the dispenser shall point in the direction of travel and the dispenser suspension lugs shall face the hub of the centrifuge. The centrifuge shall be brought up to the rotational speed required to produce 8g ± 0.5g radial acceleration. This acceleration shall be stabilized and maintained for a period of not less than one minute. To simulate side loading, the dispenser shall be rotated such that the longitudinal (roll) axis of the dispenser is normal to the arm of the centrifuge and the yaw axis of the dispenser is perpendicular to the arm of the centrifuge. The dispenser shall be locked in this position and subjected to the same acceleration for a period of not less than one minute. The dispenser shall be operable both before and after the acceleration test. No mechanical failures or malfunctions shall be evident as a result of the test. The dispenser shall be examined and tested in accordance with 4.6.3.5.

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4.6.7 Environmental tests.

4.6.7.1 During the following tests the dispenser shall be loaded with 6 each BDU-33/B Practice Bombs.

4.6.7.2 Sand and dust test. The sand and dust test shall be in accordance with Method 510, procedure I of MIL-STD-810, except that the velocity shall be increased to 2500 ± 500 feet per minute. Upon completion of the sand and dust test, the dispenser shall be examined and tested in accordance with 4.6.3.5.

4.6.7.3 Humidity test. The humidity test shall be in accordance with Method 507.1, procedure I of MIL-STD-810. The dispenser shall be operated during the fifth cycle toward the conclusion of the 6-hour high temperature period in accordance with 4.6.3.5. At the conclusion of the 240-hour period, the dispenser shall be returned to the prevailing room conditions, examined, and again tested in accordance with 4.6.3.5.

4.6.7.4 Altitude test. The altitude test shall be in accordance with Method 500.1, procedure II of MIL-STD-810. The ambient temperature of the internal test chamber shall be reduced to -43°C (-45°F) and the equipment maintained in this condition for a period of two hours. While in these conditions, the dispenser shall be tested in accordance with 4.6.3.5. The dispenser shall then be returned to normal room conditions, examined for failures, and again tested in accordance with 4.6.3.5.

4.6.7.5 High temperature test. The high temperature test shall be in accordance with Method 501.1, procedure I of MIL-STD-810. At the termination of 25 and 50 hours, while at this temperature, the dispenser shall be examined and tested in accordance with 4.6.3.5. The dispenser shall then be returned to normal room conditions, examined for failures, and again tested in accordance with 4.6.3.5.

4.6.7.6 Low temperature test. The low temperature test shall be in accordance with Method 502.1, procedure I of MIL-STD-810. At the conclusion of the exposure period, and while at this temperature, the dispenser shall be examined and tested in accordance with 4.6.3.5.

4.6.7.7 Temperature shock test. The temperature shock test shall be in accordance with Method 503.1, procedure I of MIL-STD-810. The dispenser shall be tested in accordance with 4.6.3.5, during that part of the temperature cycle when the chamber temperature is being raised from -55°C (-67°F) to 71°C (160°F). At the conclusion of the test, the dispenser shall be examined and tested again in accordance with 4.6.3.5.

4.6.7.8 Salt spray test. The salt spray test shall be in accordance with Method 509.1, procedure I of MIL-STD-810. The period shall be of 50 hours duration. At the conclusion of the test period, the salt residue shall be removed and the equipment examined for failure and tested in accordance with 4.6.3.5.

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4.6.7.9 Fungus test. The fungus test shall be in accordance with Method 508.1, procedure I of MIL-STD-810. The unit selected for this test shall not have been subjected to the salt spray test. Upon completion of the fungus test, the dispenser shall be thoroughly examined to ascertain that there is no growth of fungus present which impairs the function of the dispenser. The dispenser shall be operated in accordance with 4.6.3.5.

4.7 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Levels of preservation and packaging. Levels of preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each Dispenser, Bomb, shall be preserved in accordance with MIL-P-116, Method III. The item shall be shrouded with plastic film (polyethylene, thin gage). Minimum thickness of the film shall be 0.006 inch and shall conform to L-P-378 or equal. The shroud shall be applied in such a manner as to prevent direct entry of water and foreign matter and shall be held in place with tape, PPP-T-60 or equal. The tape shall not be allowed to come in direct contact with the surface of the dispenser.

5.1.2 Level C. Each dispenser shall be preserved and packaged in a manner which affords adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The dispenser preserved and packaged to meet the requirements of 5.1.1 shall be packed in a container conforming to MIL-C-25731, Type II, Class 2. Cushioning, blocking, bracing and anchoring shall be in accordance with MIL-STD-1186. Rough handling tests shall conform to MIL-STD-794.

5.2.2 Level B. Same as level A.

5.2.3 Level C. The dispenser shall be packed in a manner which affords adequate protection at the lowest rate against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with MIL-STD-129. In addition, shipping containers shall be marked "Reusable Container - Do Not Destroy" in an appropriate place on two sides of the container.

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6. NOTES

6.1 Intended use. The Dispenser, Bomb, Type SUU-21/A, is intended for use in dispensing six miniature practice bombs in sequence which duplicate a desired trajectory, and contain smoke charges for determining the point of impact. The SUU-21/A Dispenser is capable of dispensing BDU-23/B, BDU-33/B or MK-106 Mod 1 Practice Bombs.

6.1.1 Use with simulator. There are provisions for installing a weapons simulator in the aft section of the dispenser. The dispenser with non-selective relay box may be used with the MM-1 or SMU-10/B Simulator installed for the purpose of simulating, in flight, malfunctions of certain special weapons. The dispenser with selective option relay box cannot be used with a simulator.

6.2 Ordering data. Invitation for bids, contracts and purchase orders should specify the following:

a. Title, number and date of this specification.

b. Applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.1 Preproduction samples. It is expected that the contract or purchase order will specify that three units representative of the production dispenser will be submitted to the procuring activity for test and evaluation.

6.3 Sources. When requesting publications, refer to both title and number.

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Air Force - 79

Reviewer:
Air Force - 79

Preparing activity:
Air Force - 79

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San Antonio Air Materiel Area
Service Engineering Division (SANEOB)
Kelly AFB, Texas 78241

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENT ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

MITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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